

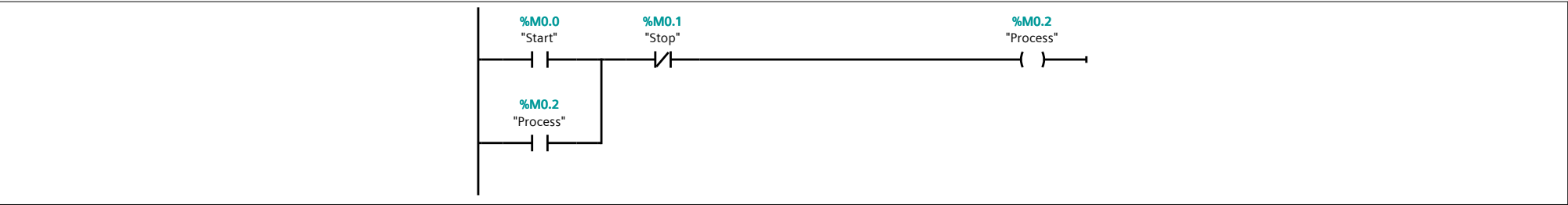
Homework Day8 / PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

Main [OB1]

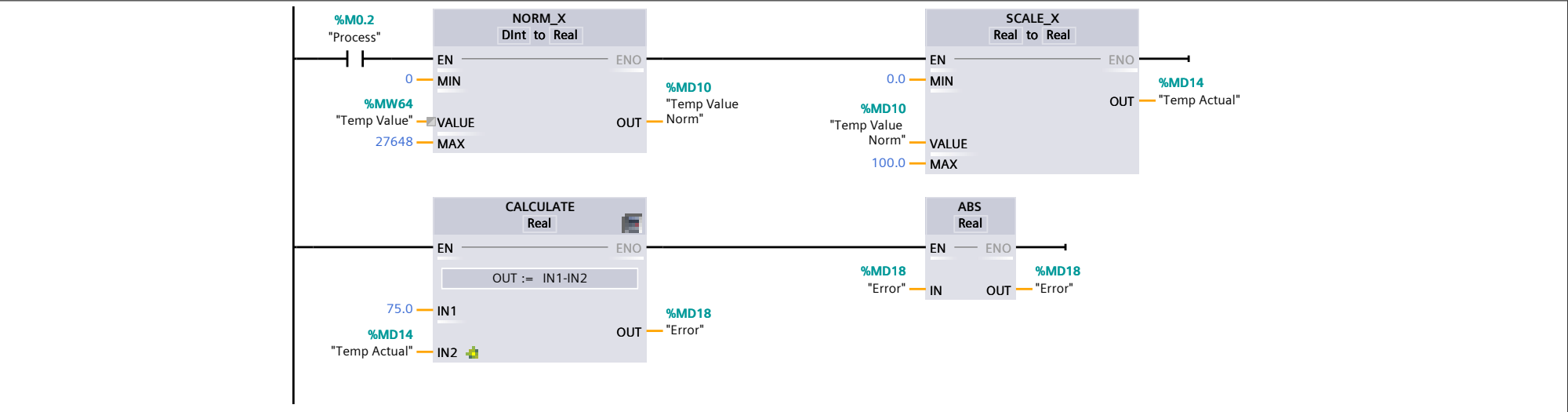
Main Properties							
General							
Name	Main	Number	1	Type	OB	Language	LAD
Numbering	Automatic						
Information							
Title	"Main Program Sweep (Cycle)"	Author		Comment	Temperature Control with PID Simulation (Math, Comparison, Timers) Scenario: A heater maintains a temperature setpoint. The current temperature is Temp_Actual (REAL) and the setpoint is Temp_Setpoint (REAL = 75.0°C). Requirements: 1. Use Math to calculate the error: Error = Temp_Setpoint - Temp_Actual. 2. Use Comparison to control the heater: • If Error > 5.0, turn heater ON 100% (continuous). If Error between 1.0 and 5.0, use a Timer to cycle the heater ON for 5 seconds, OFF for 5 seconds (50% duty cycle). • If Error < 1.0, turn heater OFF. 3. Use a Timer to sample the temperature every 1 second and store the last 5 samples (use Math to calculate the average)	Family	
Version	0.1	User-defined ID					

Main			
Name	Data type	Default value	Comment
▼ Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

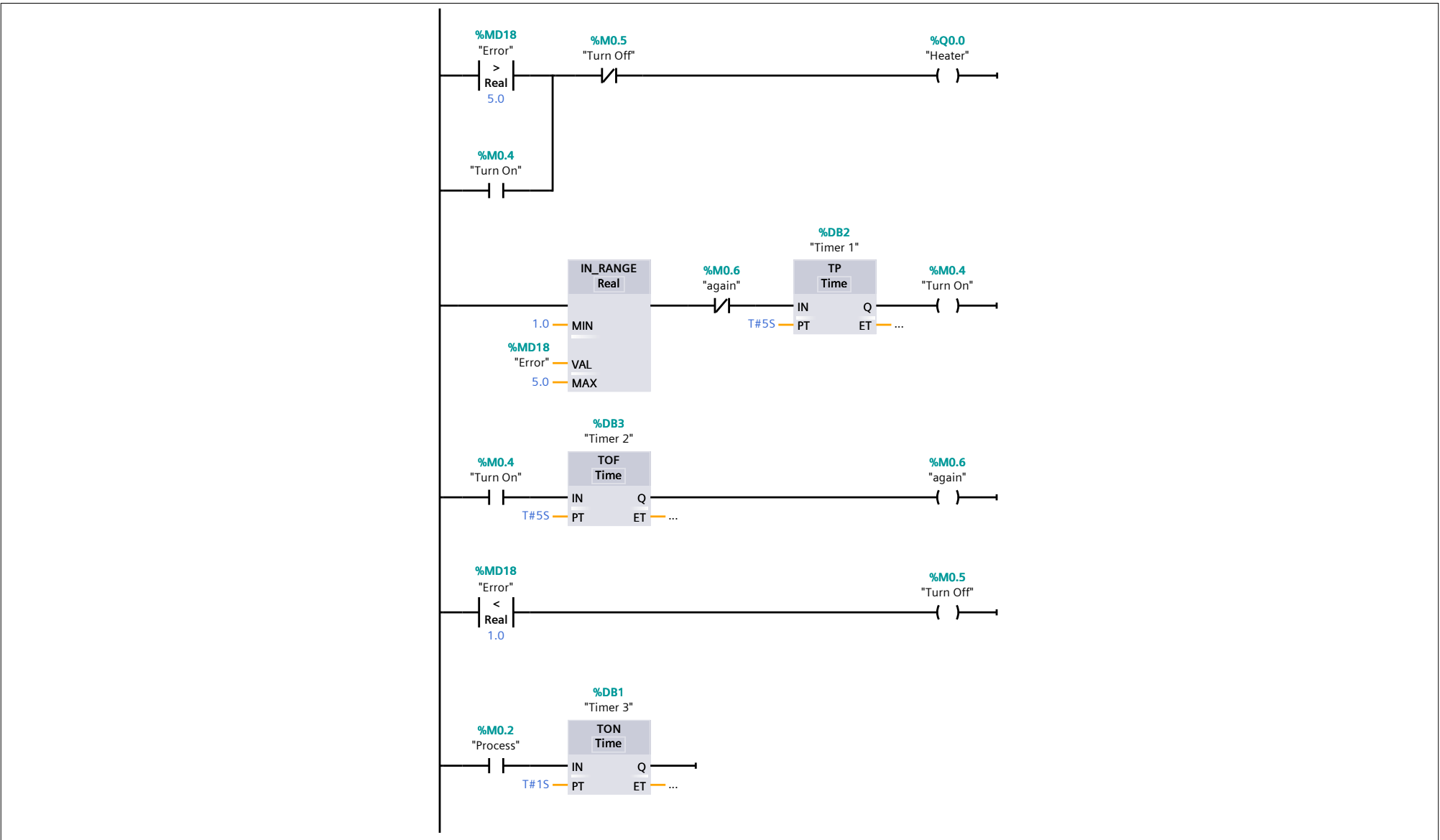
Network 1:



Network 2:



Network 3:



Network 4:

